

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Cycle Test

5 August 2024

Subject	Mathematics
Grade	12
Topic	Euclidean Geometry
Marks	50
Time	1 hour

Examiner

Dr G Nixon

Moderator

Mrs S Rowell

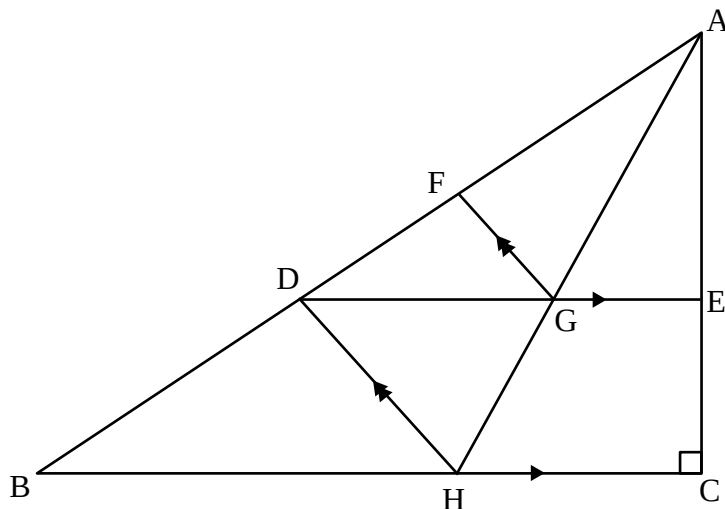
INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 3 questions and 5 pages. Please ensure that your question paper is complete.
2. Answer ALL the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

QUESTION 1

In the diagram below, $\triangle ABC \sim \triangle ADE$, $DE \parallel BC$, $FG \parallel DH$ and $\angle ACB = 90^\circ$.
Furthermore, $AE = 2 \text{ cm}$ and $EC = 1 \text{ cm}$.



1.1 Determine, with reasons, the values of the following ratios:

1.1.1 $\frac{AG}{GH}$ (2)

1.1.2 $\frac{DE}{BC}$ (2)

1.1.3 $\frac{AF}{FB}$ (4)

1.2 If $AD = 16 \text{ cm}$, determine the length of AB . (3)

1.3 It is further given that $BH = 6\sqrt{7} \text{ cm}$. Calculate the value of:

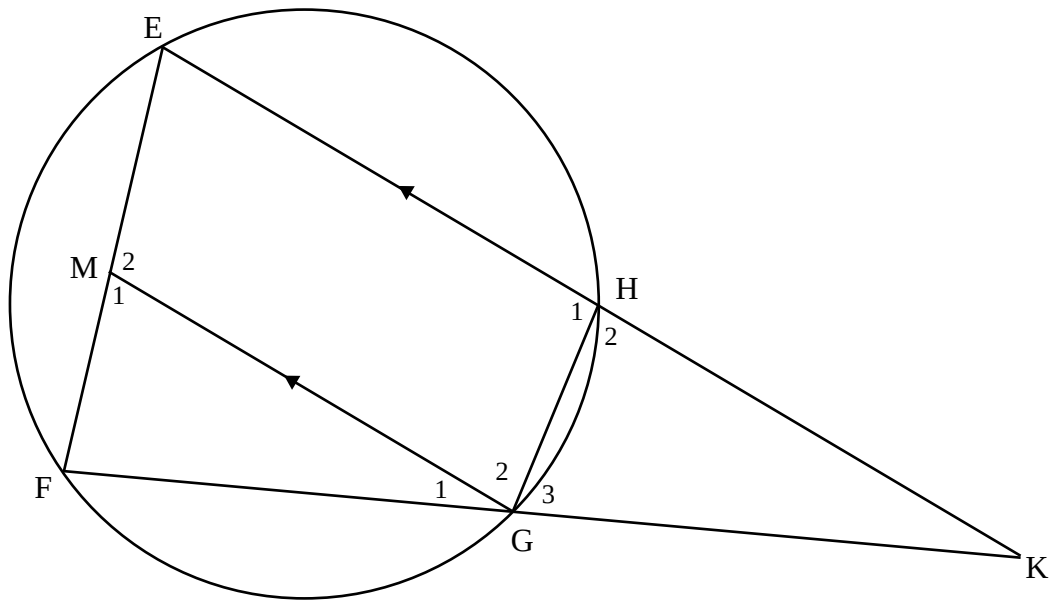
1.3.1 $\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle ABC}$ (2)

1.3.2 $\frac{\text{Area of } \triangle ABH}{\text{Area of } \triangle AHC}$ (4)

[17]

QUESTION 2

In the diagram below, E, F, G and H lie on the circumference of the circle. Chords EH and FG are produced to meet at K. M is a point on EF such that $MG \parallel EK$ and $KG = EF$.



Prove that:

$$2.1 \quad \triangle KGH \parallel \triangle KEF \quad (4)$$

$$2.2 \quad \frac{KG}{GM} = \frac{GH}{MF} \quad (5)$$

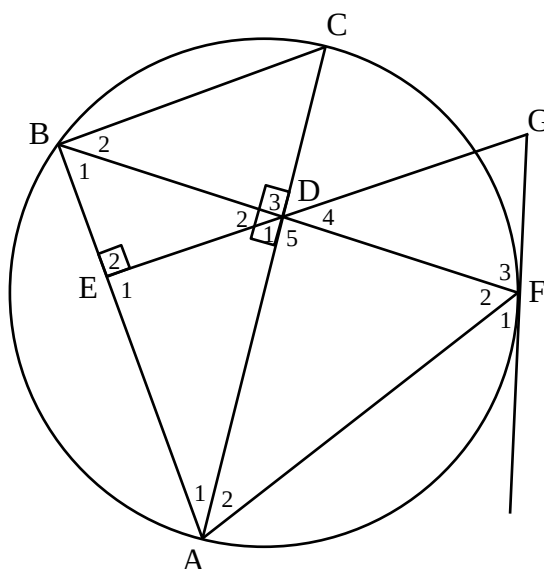
$$2.3 \quad KG^2 = EM \cdot KF \quad (2)$$

$$2.4 \quad HG \text{ is a tangent to circle } FMG. \quad (4)$$

[15]

QUESTION 3

In the diagram below, diameter AC is perpendicular to the chord BF. Line EDG is drawn such that $\hat{E}_2 = 90^\circ$. FG is a tangent to the circle at F.



- 3.1 Give a reason why $\hat{ABC} = 90^\circ$. (1)
- 3.2 Name two other triangles similar to $\triangle ABC$. Motivate, with a reason, why they are similar. (3)
- 3.3 It is further given that $AD : DC = 3 : 2$ and $DC = 2x$. Express the following in terms of x :
 - 3.3.1 AC (1)
 - 3.3.2 BD (4)
- 3.4 Prove that AC bisects $\hat{BAF}$. (3)
- 3.5 If $\hat{A}_1 = y$, determine the size of $\hat{FGD}$ in terms of y . (6)

[18]